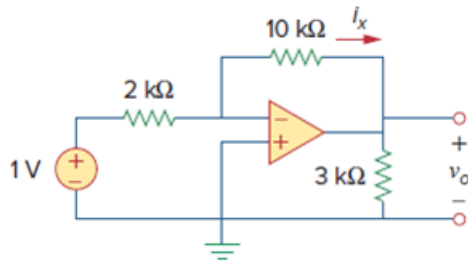


Problem

For the circuit in Fig. 5.40, voltage v_o is:

- (a) -6 V
- (b) -5 V
- (c) -1.2 V
- (d) -0.2 V

Figure. 5.40:



Step-by-step solution

Step 1 of 1

Refer to Figure 5.40 in the circuit.

Consider that the voltages at inverting and non-inverting terminals of an ideal op amp are equal due to virtual ground.

$$\begin{aligned} v_- &= v_+ \\ &= 0\text{ V} \end{aligned}$$

Apply Kirchhoff's current law at inverting node of the op amp.

$$\begin{aligned} \frac{v_- - 1\text{ V}}{2\text{ k}\Omega} + \frac{v_- - v_o}{10\text{ k}\Omega} &= 0 \\ \frac{0 - 1}{2000} + \frac{0 - v_o}{10000} &= 0 \quad (\text{since, } v_- = 0\text{ V}) \\ \frac{-v_o}{10000} &= \frac{1}{2000} \\ 2000v_o &= -10000 \\ v_o &= -\frac{10000}{2000} \\ &= -5\text{ V} \end{aligned}$$

Thus, the value of output voltage, v_o is -5 V .

Hence, the correct option is **(b)**.